

A Falsifiable Open-Quantum-Systems Model of Emotional State Evolution: A Conflict-Dependent Leggett-Garg Signature

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Note to readers (not part of the paper). This is a concept draft written to invite critique from researchers whose theories it builds on. The central quantitative claims (Sections 4 and 5) have been reproduced in independent code and, where possible, verified analytically. Sections 2 and 3 summarise the surrounding architecture briefly, since the discriminating claim does not depend on its every detail.

Abstract

We present an individual-level model of emotional (experiential) state evolution formulated entirely within the open-quantum-systems (OQS) framework, and we derive from it a single falsifiable prediction that no classical (macrorealist) model can reproduce. Each agent's affective state lives on Russell's circumplex, mapped to the Bloch equator of a qubit so that opposite emotions are orthogonal. Ambivalence appears in two distinct senses: as high von Neumann entropy of the memory state (a genuine mixture over conflicting past experiences) and, during live deliberation, as a coherent superposition of competing outcomes (a pure, oscillating state); the latter is where the quantum signature we test resides. Perception, Husserlian retention, autobiographical memory, and two socially bounded memory fields interfere in a five-source superposition; emotional intensity is defined as interference visibility. A single quantity, the exact qubit trace distance between perception and memory read through the Helstrom bound, drives dual-process route selection (recognition, surprise, deliberation) with thresholds emergent from each agent's own history and no scale constants. Deliberation is modelled as an open-system quantum trajectory on a qutrit whose pointer states are *perception*, *conflict*, and *limbo* (indecision as a located, non-empty outcome); decision time is emergent from an environmental readout rate rather than an iteration cap. The conflict that drives this trajectory is endogenous, arising from the angle between perception and expectation (a protentional term in the Hamiltonian).

Our central claim is a discriminating one. The ambivalence dynamics underlying deliberation, when probed by ideal sequential measurements, violate the Leggett-Garg inequality, and the degree of violation grows monotonically with the conflict angle: the third-order Leggett-Garg parameter K_3 rises from ≈ 1.06 at weak conflict to ≈ 1.39 at maximal conflict under realistic environmental decoherence (bounded above by the qubit maximum 1.5). Because every macrorealist model satisfies $K_3 \leq 1$ by theorem, the *shape* of the predicted $K_3(\text{conflict})$ curve, not the mere fact of violation, is a signature that classical affect models cannot produce. We propose a concrete human experiment to test it, and we report a classical Markov-jump twin that reproduces the

model’s outcome and decision-time distributions yet remains bounded by $K_3 \leq 1$, isolating quantum coherence as the load-bearing ingredient.

As an empirical contact point that we *can* test with existing data, the model predicts that event-triggered ambivalence relaxes monotonically to a personal baseline (the “limbo after-effect”). In 222 participants across two experience-sampling datasets this holds without exception (AR(1) persistence $\varphi < 1$ for 100% of participants; monotone event-triggered decay). We report throughout what has been verified in code, what remains open, and one earlier overclaim that replication retracted.

Keywords: quantum cognition, open quantum systems, Leggett-Garg inequality, affect dynamics, circumplex, ambivalence, decision times

1. Introduction and positioning

People recognise some situations and respond on learned autopilot, are caught off guard by others, and must deliberate over a third kind, sometimes without reaching a resolution at all. This paper develops a formal model in which these response modes, including the failure to decide, emerge from open-quantum-systems dynamics rather than being programmed, and in which the deliberation process makes a prediction that separates a genuinely quantum-like account from any classical one.

We work in the quantum-cognition tradition (Busemeyer & Bruza, 2012; Pothos & Busemeyer, 2022), which applies the mathematics of quantum theory to cognition without asserting quantum processes in the brain, and specifically in the open-quantum-systems strand developed for the cognitive and social sciences by Khrennikov (2023). Our aim is to contribute where we believe this literature is thin, and to be explicit about our debt to it.

Relative to existing work, the contribution is narrow and, we hope, sharp:

- Most quantum-cognition models target judgment and decision phenomena (order effects, the conjunction fallacy, probability updating) and are evaluated by *fitting* human choice data. We target emotional *state evolution*, and instead of a fit we offer a *discriminating prediction*: a measurable regime where quantum-like and classical accounts diverge.
- Khrennikov’s OQS programme supplies the formal apparatus (density matrices, Lindblad dynamics, decoherence as environmental monitoring). We instantiate it concretely for affect and show that the resulting ambivalence dynamics carry a Leggett-Garg signature.
- The closest work on emotion specifically (Gnidko & Vasilenko, 2026) models quantum-like collective emotion in social networks with a deliberately thin formalism and no individual-level falsifiable prediction. Our model is complementary: a fully QM-consistent individual architecture (valid density matrices throughout, exact closed-form qubit trace distance and fidelity, open-system relaxation) from which social structure emerges, and which stakes a testable claim.

In one sentence: *we present the first individual-level OQS model of emotional dynamics that yields a falsifiable, conflict-dependent Leggett-Garg signature no macrorealist model can reproduce.* Everything else in the architecture, the circumplex qubit, the

five-source interference, the emergent thresholds, the decaying character matrix, the emergent group realities, is the machinery that makes this a coherent affective model rather than an isolated toy; the Leggett-Garg prediction is where it earns the label “quantum.”

The only deliberate departure from standard quantum mechanics is the collapse rule for state selection: we use projective normalisation rather than the Born rule, stated as a postulate variant, not an approximation. We report failures alongside successes, including one retracted overclaim (Section 5).

2. State space and superposition

An emotional state is a direction on Russell’s (1980) circumplex (valence $\times$ arousal). We map the emotion circle to the Bloch equator of a qubit by setting the state angle to half the emotion angle. This matters: under the full angle a state and its negation share a density matrix, so an agent oscillating between joy and anger would read as maximally coherent; under the half-angle (Bloch) convention opposite emotions are orthogonal and ambivalence over remembered experience registers as high von Neumann entropy and low purity. This entropy reading concerns the memory ensemble, a genuine mixture over conflicting past states, and should not be confused with the moment-to-moment ambivalence of active deliberation, which is a coherent superposition (Section 3) and is where the quantum signature lives. A single pure qubit is never itself ambivalent; it is simply some emotional direction. Ambivalence is therefore either the mixedness of memory or the coherence of an unresolved deliberation, two different things that this paper keeps apart. Memory is summarised by a density matrix ρ ; from it we take entropy S , purity P , and Bloch length $m = \sqrt{2P-1}$. All matrix quantities use exact closed qubit forms (fidelity $F(\rho, \sigma) = \text{tr}(\rho\sigma) + 2\sqrt{(\det \rho \det \sigma)}$; trace distance $T = \frac{1}{2}\|\rho - \sigma\|_1$), verified numerically against eigenvalue references (max deviation $\sim 10^{-15}$).

At each perception event five sources interfere,

$$\Psi = A_w + \gamma \cdot A_r + \delta \cdot A_h + \beta_g \cdot A_g + \beta_e \cdot A_e \text{ (projectively normalised),}$$

with A_w the current perception, A_r the Husserlian retention (the immediately preceding collapsed state, weighted by $\gamma = P \cdot F$), A_h the autobiographical attractor (weighted by $\delta = m$), and A_g, A_e the group and external memory fields with weights bounded by $\beta \leq P/\sqrt{N}$ at the natural group sizes. Emotional intensity is defined as interference visibility, $I = |\Psi_{\text{prenorm}}| / \Sigma|\text{components}| \in [0,1]$: aligned sources reinforce, opposed sources damp, so ambivalence is destructive interference and the collapsed state carries reduced intensity. (Under the half-angle convention, maximal ambivalence damps toward the orthogonal “tense” direction rather than to zero.) Route selection among recognition, surprise, and deliberation uses thresholds computed from each agent’s own history, its dispositional self-fluctuation and the adaptive quantile of its own experienced distances, with no scale constants. Character is an initial density matrix ρ_0 with dynamically decaying weight; group affinity is the trace distance between dispositional density matrices. These components are summarised only briefly here; they are not the focus of this concept.

3. Deliberation as an open-system qutrit trajectory

Deliberation (route B) is the regime in which perception is neither recognised nor surprising. We model it as a quantum trajectory of an open three-level system whose pointer basis is

$|0\rangle$ = perception pool, $|1\rangle$ = conflict pool (the opposed, expectation-driven emotion), $|2\rangle$ = limbo (torn, unresolved).

The Hamiltonian drives coherent exchange between perception and conflict and feeds the limbo channel; both couplings are **endogenous**, derived from the conflict inherent in the input rather than injected as free stimulus. The conflict strength follows from the angle $\Delta\theta$ between the perception and the expectation (retention plus disposition): $f = (1 - \cos \Delta\theta)/2$. This is the protentional element of the model: expectation is not a passive prior but an active term in the generator of the dynamics. During deliberation the state is thus a coherent superposition of the three pointers (a pure, low-entropy, oscillating state), and it is this coherence, not the entropy of memory, that produces the Leggett-Garg signature of Section 4. This aligns with the open-quantum-systems account of ambivalence in decision making of Rosner et al. (2022), whose qutrit basis $|\text{yes}\rangle$, $|\text{undecided}\rangle$, $|\text{no}\rangle$ our perception, conflict, and limbo pointers mirror; their $|\text{undecided}\rangle$ is the counterpart of our limbo, and in both accounts it is decoherence, the growth of entropy, that marks the resolution of ambivalence rather than its presence.

The environment monitors the system as a Poisson process at rate r ; the first monitoring event is the collapse, with Born probabilities over the three pointers. Two consequences distinguish this from the earlier (v1) formulation:

1. **Decision time is emergent.** It is the time of the first readout, a distributed quantity, with no maximum-iteration cap and no fixed threshold. The rate r separates *timing* (when a decision is reached) from *conflict* (what is decided).
2. **Indecision is a located outcome, not a truncation.** “Limbo” is a genuine third result, a non-empty, localised state, rather than a run that failed to converge. Its after-effect is carried on a separate, classical “groundedness” layer that relaxes monotonically to a trait baseline (Section 5), which is what we test against experience-sampling data.

This replaces the v1 deliberation rule (an iteration loop truncated at 50,000 steps and flagged as indecisive), which we now regard as an artefact of a coincidence-waiting stopping process. The qutrit trajectory resolves the stopping-rule question that the earlier draft posed as open, and it lifts route-B dynamics above the single-qubit limitation (the emotion state space elsewhere remains a qubit; the dominance/PAD dimension is still absent, a limitation we retain).

4. The Leggett-Garg prediction

The perception-conflict oscillation that underlies deliberation is a coherent two-level dynamics subject to environmental dephasing. This is exactly the setting for a Leggett-Garg test of macrorealism. Define the dichotomic observable $Q = +1$ (leaning toward the perceived emotion) versus $Q = -1$ (leaning toward the expected/conflict emotion), and the third-order parameter

$$K_3 = C(0, \tau) + C(\tau, 2\tau) - C(0, 2\tau),$$

with C the two-time correlation from sequential projective measurements. Every model satisfying macrorealism and non-invasive measurability obeys $K_3 \leq 1$; a coherent qubit can reach $K_3 = 1.5$.

Result (reproduced in independent code). With the ambivalence Hamiltonian $H = (\Delta/2)\sigma_x$ and z -basis dephasing at rate r , K_3 exceeds 1 across the whole realistic range, and its value grows monotonically with the conflict angle. At $r = 0.5$ (representative decoherence), maximised over the measurement interval τ :

conflict angle	Δ	max K_3
30°	0.16	1.06
60°	0.60	1.22
90°	1.20	1.32
135°	2.05	1.38
180°	2.40	1.39

The decoherence-free calibration point ($r = 0, 180^\circ$) returns the qubit maximum $K_3 = 1.500$, as it must; increasing r degrades but does not destroy the violation ($K_3 = 1.12$ even at $r = 4$). These figures were verified by two independent implementations, a Bloch-equation integrator and a density-matrix superoperator with explicit projective sequential measurements, agreeing to the third decimal.

What the prediction is, and is not. The bare fact that a coherent qubit violates Leggett-Garg is textbook and, on its own, no contribution; so is the observation that the violation grows with the coherent coupling, which is generic to any two-level system. The claim we stake is sharper: the model fixes the *specific functional shape* of the $K_3(\text{conflict})$ curve, because the coupling is not a free knob but a derived quantity. The conflict angle θ between percept and expectation enters through their overlap as $\Delta(\theta) \propto (1 - \cos \theta)/2$, a form dictated by the geometry and not fitted to produce a violation. Composed with the Rabi-Leggett-Garg response this yields a particular curve (shallow at small conflict, saturating toward maximal conflict) that differs in shape from what a generic coherent model, for instance one with coupling linear in θ , would give. The prediction is therefore tested as a *trend across several conflict conditions*, by asking whether the measured curve has this parameter-free shape, not by asking whether a single violation exceeds 1. Because macrorealism caps K_3 at 1 at every conflict angle, and a generic coherent alternative predicts a differently shaped curve, this shape is a signature that neither a classical affect model nor an arbitrary quantum-like one reproduces.

How distinguishable the shape is, quantitatively. We tested this claim directly. Against a conflict-independent (constant) coupling, the null that conflict does not matter, the curve is clearly separable at a modest sample of about 100 participants per conflict level. Against a generic monotone (linear) coupling the full-curve difference is small, and separating the two across the whole range would need roughly 800 participants per level. The discriminating feature is the onset: $(1 - \cos \theta)/2$ rises quadratically, so the model predicts almost no violation at small conflict ($K_3 \approx 1.005$ at 15°) whereas a linear coupling rises immediately. A test of the shape should therefore sample densely at small conflict angles, where the accounts diverge, rather than only at

large ones. We state this plainly because it bounds the claim honestly: the linear alternative is itself a coherent, non-macrorealist model, so this qualification concerns only which coherent coupling is at work and leaves the quantum-versus-classical contrast (a flat $K_3 \leq 1$ for any macrorealist account versus a rising curve) untouched. Two honest constraints travel with it: (i) K_3 is defined on the perception-conflict distinction (the full three-level case, with the limbo channel active, is treated in the robustness check below); (ii) it uses ideal projective measurements at experimenter-chosen times, with environmental dephasing r running throughout, the standard Leggett-Garg setup, not the model’s own Poisson-readout collapse. The prediction is about the dynamics an experimenter would find on probing the system, not a claim that the model’s internal decision process performs Leggett-Garg measurements on itself.

Isolating the quantum ingredient. To show that coherence, and not the three-state structure or the rates, does the work, we built a classical Markov-jump twin: the same three pointer states (perception, conflict, limbo), the same environmental readout rate, and transition rates calibrated to the model’s own outcome distribution, but evolving classical populations only, with no coherences. Under the identical Leggett-Garg procedure the twin reproduces the model’s decision-time distribution exactly (both use Poisson readout) and its outcome distribution well in the realistic regime (mean cell deviation ≈ 0.13 for $r \geq 0.5$), yet its K_3 never exceeds 1 (range 0.989 to 1.000 across all conflict angles and decoherence rates). The same holds for the tightest control, the model’s own dynamics with the off-diagonal density-matrix elements removed at each step, which returns $K_3 = 1.000$ exactly (stripping coherence both freezes the population transfer and removes the violation). Only the coherent model reaches $K_3 > 1$, rising with conflict. This is guaranteed by structure, not luck: for the classical decay the parameter reduces to $K_3 = 2u - u^2 \leq 1$ ($u = e^{-\tau}$), with equality only in the limit. One informative discrepancy: at very low decoherence ($r = 0.25$) the classical twin over-leaks toward limbo relative to the quantum model (at 180° : 30/19/51 versus 54/23/23), because the coherent “holding” of the perception-conflict balance is itself a coherence effect the classical model cannot mimic, additional evidence that the quantum structure is doing visible work even at the level of outcome frequencies. The classical-twin and decohered-control results were reproduced in independent code and verified analytically.

Robustness (three adversarial checks). We attacked the prediction on the three points a referee is most likely to press, and it held on all three; each was reproduced in independent code and, where possible, verified analytically. First, the two-level restriction: recomputing K_3 on the full three-level open system, with the limbo channel active and a coarse-grained observable (perception versus not-perception), leaves the violation intact and if anything slightly larger. At the model’s own limbo-coupling strength, $\max K_3 = 1.39$ at maximal conflict and stays above 1 at every conflict angle and coupling tested. Second, arbitrary constants: $\max K_3$ is time-scale invariant and depends only on the dimensionless ratio Δ/r , so the chosen magnitudes do not matter; across the full character-emergent range of that ratio the curve stays above 1 (minimum ≈ 1.01 in the weakest corner) and monotone in conflict, not confined to a tuned window. Third, the collapse postulate: because the Leggett-Garg measurement uses orthogonal projectors, the model’s projective-normalisation rule reduces there exactly to the Born rule (the two give identical K_3 to machine precision), and $\max K_3$ decreases monotonically with the dephasing rate r , so the environment degrades the violation rather than

manufacturing it. The projective-normalisation postulate governs emotional intensity, not the dichotomic outcome, and so does not touch the prediction.

A proposed test in humans, in an existing paradigm. The prediction can be tested with the sequential change-judgment paradigm that Waddup, Yearsley, Blasiak & Pothos (in press) developed for exactly this class of temporal-Bell (Leggett-Garg) test in cognition. Rather than asking for the value of Q at each time, which would disturb the state, that paradigm asks only whether a dichotomic judgment changed between two timepoints, which is how they control the measurement-disturbance (clumsiness) loophole. In the change-count form (Atmanspacher & Filk, 2010), macrorealism requires $N_{13} \leq N_{12} + N_{23}$, where N_{ij} is the probability that Q differs between times i and j . This is exactly our K_3 re-expressed: through $C_{ij} = 1 - 2 \cdot N_{ij}$ one has $K_3 = 1 + 2 \cdot (N_{13} - N_{12} - N_{23})$, an identity we confirmed against the K_3 computation to machine precision.

Expressed in that observable, the model predicts a positive, conflict-dependent margin $N_{13} - (N_{12} + N_{23})$, at $r = 0.5$ and the K_3 -maximising interval:

conflict angle	N_{12}	N_{23}	N_{13}	$N_{13} - (N_{12} + N_{23})$
30°	0.05	0.05	0.13	+0.03
90°	0.18	0.18	0.52	+0.16
180°	0.22	0.22	0.63	+0.20

The point of contact is precise. Waddup et al. measured this margin at a single operating point and reported a marginal violation; the model turns it into a graded prediction, in which the margin grows monotonically with the conflict between percept and expectation. That shape is what a macrorealist account cannot produce and what a generic “any dynamical system can violate Leggett-Garg” argument does not predict. We do not have the facilities to run the experiment and offer it as a registered-report-shaped extension of an established paradigm. (Full protocol in the accompanying document.)

5. Empirical contact point: the limbo after-effect

The one prediction we can already confront with data concerns the after-effect of unresolved deliberation. In the model, a limbo outcome raises a slow “groundedness” state that relaxes monotonically toward a personal baseline, a monostable trait, with no bistability or scarring. This implies that in daily life, episodes of elevated ambivalence should decay back to a personal baseline rather than persist or spiral.

We tested this on two experience-sampling datasets (Bringmann 2013, $n = 127$; Bringmann 2016, $n = 95$; 222 participants total), using $\min(\text{positive}, \text{negative})$ affect as an ambivalence proxy. The AR(1) persistence of ambivalence is below 1 for 100% of participants (median $\phi = 0.31$ and 0.23 ; no participant showed $\phi \geq 1$, i.e. no spiral), and event-triggered ambivalence decays monotonically back to baseline over subsequent measurements (deviation from personal mean at lags +1 to +5: 0.056, 0.036, 0.023, 0.013, 0.007 in Bringmann 2013; a comparable monotone decay in 2016). This is qualitative support: the ambivalence proxy is noisy (a known deflation of ϕ), and the finding speaks to the *shape* of recovery, not to a quantitative rate. It is nonetheless a genuine,

pre-registered, reproducible contact point between the model’s indecision mechanism and human affect dynamics.

A second, weaker structural contact point: because one quantity ($\gamma = P \cdot F$) drives both retention and attractor pinning, the model predicts emotional inertia and (low) affect variability to be coupled. Against six experience-sampling datasets (689 participants) the human coupling is $r \approx +0.30$; the model’s is $r \approx +0.55$ (replication range 0.37 to 0.71), the same sign and order, model coupling stronger. We initially reported an exact match (+0.29) from a single run; replication exposed this as sampling luck and the claim was retracted. We report both values and treat this only as structural, non-diagnostic agreement.

AI use statement

This model originates in the author’s own way of thinking. Working outside academia, without a stake in any particular school, the author is a visual thinker with a background in creative digital media and venture building, and approached emotion, ambivalence, and indecision from a phenomenological intuition, following it wherever it led, including to the unconventional idea that the mathematics of open quantum systems might capture the structure of experiential dynamics. The conceptual architecture, the design decisions, and the framing of what would count as a discriminating test are the author’s.

The author’s mode of control was visual rather than mathematical. Alongside the model the author built an interactive simulation that renders each mechanism and its emergent behaviour in real time; this visual representation is the instrument through which the author judged whether the model behaved correctly and decided what to change next. Watching this simulation is also how the wider behaviour was explored: running several agents together, for instance, shows populations spontaneously converging on a shared reality or splitting into lasting subgroups. Such observations are offered as an illustration of this visual, exploratory method rather than as validated claims. Artificial intelligence (Anthropic’s Claude) served as a mathematical instrument, not a source of ideas: it translated the author’s intuitions into the quantum-mechanical formalism, implemented and audited the linear algebra, and ran the analyses reported here. The author does not read the code and makes no claim to have checked the mathematics line by line. Instead the formal machinery is archived as open, executable, and independently reproducible, and the central results reported here were reproduced in independent implementations. The division of labour is deliberately transparent: the ideas, and their validation through the model’s behaviour, are the author’s; the formal derivations were built with machine assistance and are open to anyone who wishes to check them. The author directed the work throughout and takes full responsibility for all content.

Data and code availability

Everything needed to reproduce this concept is openly available in the public repository *quantum-experiential-dynamics*: the interactive model as a single self-contained HTML file (viewable in the browser at willemvanderkrieken.github.io/quantum-experiential-dynamics/model/bewustzijn-model-v6.1.html), the Leggett-Garg analysis, the classical-

twin and decohered-control comparison, the three robustness tests, the experience-sampling analysis, and the full development audit. The work is archived with a timestamped record at Zenodo (doi.org/10.5281/zenodo.21499287). Third-party datasets are referenced, not redistributed.